The due date for submitting this assignment has passed.
Due on 2026-03-04, 23:59 IST.

You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-03-04, 09:28 IST

Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.

Multiple Select Questions (MSQ)

1) Which of the following sets with the given addition and scalar multiplication operations (scalars are real numbers in every case) do not form vector spaces? ***1 point***

- [x] $V_1 = \{(x,y) | x,y \in \mathrm{R}\}$
  $Addition: (x_1,y_1) + (x_2,y_2) = (x_1 + x_2, 1);\ (x_1,y_1),(x_2,y_2) \in V_1$
  $Scalar\ multiplication: c(x,y) = (cx, 1);\ (x,y) \in V_1,\ c \in \mathrm{R}$

- [x] $V_2 = \{(x,y) | x,y \in \mathrm{R}\}$
  $Addition: (x_1,y_1) + (x_2,y_2) = (x_1 + x_2, y_1 + y_2);\ (x_1,y_1),(x_2,y_2) \in V_2$
  $Scalar\ multiplication: c(x,y) = (cx, 0);\ (x,y) \in V_2,\ c \in \mathrm{R}$

- [x] $V_3 = \{(x,y) | x,y \in \mathrm{R}\}$
  $Addition: (x_1,y_1) + (x_2,y_2) = (x_1 + x_2 + y_1 + y_2, x_1 + x_2 + y_1 + y_2);$
  $(x_1,y_1),(x_2,y_2) \in V_3$
  $Scalar\ multiplication: c(x,y) = (cx, cy);\ (x,y) \in V_3,\ c \in \mathrm{R}$

- [ ] $V_4 = \{(x,y,z) | x,y,z \in \mathrm{R}, x + y = z\}$
  $Addition: (x_1,y_1,z_1) + (x_2,y_2,z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2);$
  $(x_1,y_1,z_1),(x_2,y_2,z_2) \in V_4$
  $Scalar\ multiplication: c(x,y,z) = (cx, cy, cz);\ (x,y,z) \in V_4,\ c \in \mathrm{R}$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$V_1 = \{(x,y) | x,y \in \mathrm{R}\}$